GLUE COMPACTIFICATIONS ACROSS AN ELEMENTARY DISTINGUISHED SQUARE OF ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. The author writes X_1 minus Zbar_1,2 union Tbar without parentheses at3443. The surrounding3460–3461 fixes the intended removal of the union of both closed sets; the original display is retained.

Editorial note. At3477–3480 the closed map is understood on the closure of the lifting locus, as explicitly named in3478 and3498; this is the proper closure used to lift a boundary geometric point.

Editorial note. The old dump was truncated near3475. The task retains the full original proof through3538, including the decisive valuation-extension close.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Algebraic spaces, elementary distinguished squares, blowups and strict transforms, etale pushouts, properness and its refined valuative criterion, and common blowup domination. The required local boundary-separation and finite-flat blowup lemmas are retained.

Source reference. Original theorem 04S6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6209–6209. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0DVI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17363–17363. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 084Q: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10330–10330. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 086F: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10390–10390. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original definition 086B: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10386–10386. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Date: stacks-project:spaces-pushouts.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 03XQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5157–5157. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0864: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10379–10379. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0855: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10345–10345. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0853: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10343–10343. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0A4G: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12774–12774. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 087F: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10425–10425. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 086D: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10388–10388. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 087E: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10424–10424. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0878: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10418–10418. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0AGD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13191–13191. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two

concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 05WR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7595–7595. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 04GG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5832–5832. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 03KS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4688–4688. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 082Z: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10269–10269. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 088G: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10461–10461. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 03KN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4683–4683. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 04NX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6095–6095. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 03KA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4670–4670. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0ASK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13583–13583. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 088Q: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10470–10470. Stable Stacks tag for an expressly

allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0A3W: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12754–12754. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 03KU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4690–4690. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 089G: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10496–10496. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 0F3T: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18878–18878. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Lemma 0.1. *Let B be an algebraic space of finite type over $\mathbf{Z}$. Let U be an algebraic space of finite type and separated over B. Let $(U_2 \subset U, f : U_1 \to U)$ be an elementary distinguished square. Assume U_1 and U_2 have compactifications over B and $U_1 \times_U U_2 \to U$ has dense image. Then U has a compactification over B.*

This section is the analogue of More on Flatness, Section 0F3T. The theorem in this section is the main theorem in [1].

Let B be a quasi-compact and quasi-separated algebraic space over some base scheme S. We will say an algebraic space X over B *has a compactification over* B or *is compactifyable over* B if there exists a quasi-compact open immersion $X \to \overline{X}$ into an algebraic space $\overline{X}$ proper over B. If X has a compactification over B, then $X \to B$ is separated and of finite type. The main theorem of this section is that the converse is true as well.

Lemma 0.2. *Let S be a scheme. Let V, U be algebraic spaces over S. Let $V' \subset V$ be an open subspace and let $f' : V' \to U$ be a separated étale morphism of algebraic spaces over S. Then there exists a pushout*

$$\begin{array}{ccc} V' & \longrightarrow & V \\ \downarrow & & \downarrow f \\ U & \longrightarrow & W \end{array}$$

in the category of algebraic spaces over S and moreover $(U \subset W, f : V \to W)$ is an elementary distinguished square.

Proof. We are going to construct W as the quotient of an étale equivalence relation R on $U \amalg V$. Such a quotient is an algebraic space for example by Bootstrap, Theorem 04S6. Moreover, the proof of Lemma 0DVI tells us to take

$$R = U \amalg V' \amalg V' \amalg V \amalg (V' \times_U V' \setminus \Delta_{V'/U}(V'))$$

Since we assumed $V' \to U$ is separated, the image of $\Delta_{V'/U}$ is closed and hence the complement is an open subspace. The morphism $j : R \to (U \amalg V) \times_S (U \amalg V)$ is given by

$$u,\ v',\ v',\ v,\ (v'_1, v'_2) \mapsto (u, u),\ (f'(v'), v'),\ (v', f'(v')),\ (v, v),\ (v'_1, v'_2)$$

with obvious notation. It is immediately verified that this is a monomorphism, an equivalence relation, and that the induced morphisms $s, t : R \to U \amalg V$ are étale. Let $W = (U \amalg V)/R$ be the quotient algebraic space. We obtain a commutative diagram as in the statement of the lemma. To finish the proof it suffices to show that this diagram is an elementary distinguished square, since then Lemma 0DVI implies that it is a pushout. Thus we have to show that $U \to W$ is open and that f is étale and is an isomorphism over $W \setminus U$. This follows from the choice of R; we omit the details. □

Lemma 0.3. *Let S be a scheme. Let $X \to Y$ be a morphism of algebraic spaces over S. If $(U \subset X, f : V \to X)$ is an elementary distinguished square such that $U \to Y$ and $V \to Y$ are separated and $U \times_X V \to U \times_Y V$ is closed, then $X \to Y$ is separated.*

Proof. We have to check that $\Delta : X \to X \times_Y X$ is a closed immersion. There is an étale covering of $X \times_Y X$ given by the four parts $U \times_Y U$, $U \times_Y V$, $V \times_Y U$, and $V \times_Y V$. Observe that $(U \times_Y U) \times_{(X \times_Y X), \Delta} X = U$, $(U \times_Y V) \times_{(X \times_Y X), \Delta} X = U \times_X V$, $(V \times_Y U) \times_{(X \times_Y X), \Delta} X = V \times_X U$, and $(V \times_Y V) \times_{(X \times_Y X), \Delta} X = V$. Thus the assumptions of the lemma exactly tell us that Δ is a closed immersion. □

Lemma 0.4. *Let S be a scheme. Let X be a quasi-compact and quasi-separated algebraic space over S. Let $U \subset X$ be a quasi-compact open.*

1. *If $Z_1, Z_2 \subset X$ are closed subspaces of finite presentation such that $Z_1 \cap Z_2 \cap U = \emptyset$, then there exists a U-admissible blowing up $X' \to X$ such that the strict transforms of Z_1 and Z_2 are disjoint.*
2. *If $T_1, T_2 \subset |U|$ are disjoint constructible closed subsets, then there is a U-admissible blowing up $X' \to X$ such that the closures of T_1 and T_2 are disjoint.*

Proof. Proof of (1). The assumption that $Z_i \to X$ is of finite presentation signifies that the quasi-coherent ideal sheaf $\mathcal{I}_i$ of Z_i is of finite type, see Morphisms of Spaces, Lemma 084Q. Denote $Z \subset X$ the closed subspace cut out by the product $\mathcal{I}_1\mathcal{I}_2$. Observe that $Z \cap U$ is the disjoint union of $Z_1 \cap U$ and $Z_2 \cap U$. By Divisors on Spaces, Lemma 086F there is a $U \cap Z$-admissible blowup $Z' \to Z$ such that the strict transforms of Z_1 and Z_2 are disjoint. Denote $Y \subset Z$ the center of this blowing up. Then $Y \to X$ is a closed immersion of finite presentation as the composition of $Y \to Z$ and $Z \to X$ (Divisors on Spaces, Definition 086B and Morphisms of Spaces, Lemma 03XQ). Thus the blowing up $X' \to X$ of Y is a U-admissible blowing up. By general properties of strict transforms, the strict transform of Z_1, Z_2 with respect to $X' \to X$ is the same as the strict transform of Z_1, Z_2 with respect to $Z' \to Z$, see Divisors on Spaces, Lemma 0864. Thus (1) is proved.

Proof of (2). By Limits of Spaces, Lemma 0855 there exists a finite type quasi-coherent sheaf of ideals $\mathcal{J}_i \subset \mathcal{O}_U$ such that $T_i = V(\mathcal{J}_i)$ (set theoretically). By Limits of Spaces, Lemma 0853 there exists a finite type quasi-coherent sheaf of ideals $\mathcal{I}_i \subset \mathcal{O}_X$ whose restriction to U is $\mathcal{J}_i$. Apply the result of part (1) to the closed subspaces $Z_i = V(\mathcal{I}_i)$ to conclude. □

Lemma 0.5. *Let S be a scheme. Let $f : X \to Y$ be a proper morphism of quasi-compact and quasi-separated algebraic spaces over S. Let $V \subset Y$ be a quasi-compact open and $U = f^{-1}(V)$. Let $T \subset |V|$ be a closed subset such that $f|_U : U \to V$*

is an isomorphism over an open neighbourhood of T in V. Then there exists a V-admissible blowing up $Y' \to Y$ such that the strict transform $f' : X' \to Y'$ of f is an isomorphism over an open neighbourhood of the closure of T in $|Y'|$.

Proof. Let $T' \subset |V|$ be the complement of the maximal open over which $f|_U$ is an isomorphism. Then T', T are closed in $|V|$ and $T \cap T' = \emptyset$. Since $|V|$ is a spectral topological space (Properties of Spaces, Lemma 0A4G) we can find constructible closed subsets T_c, T'_c of $|V|$ with $T \subset T_c$, $T' \subset T'_c$ such that $T_c \cap T'_c = \emptyset$ (choose a quasi-compact open W of $|V|$ containing T' not meeting T and set $T_c = |V| \setminus W$, then choose a quasi-compact open W' of $|V|$ containing T_c not meeting T' and set $T'_c = |V| \setminus W'$). By Lemma 0.4 we may, after replacing Y by a V-admissible blowing up, assume that T_c and T'_c have disjoint closures in $|Y|$. Let Y_0 be the open subspace of Y corresponding to the open $|Y| \setminus \overline{T}'_c$ and set $V_0 = V \cap Y_0$, $U_0 = U \times_V V_0$, and $X_0 = X \times_Y Y_0$. Since $U_0 \to V_0$ is an isomorphism, we can find a V_0-admissible blowing up $Y'_0 \to Y_0$ such that the strict transform X'_0 of X_0 maps isomorphically to Y'_0, see More on Morphisms of Spaces, Lemma 087F. By Divisors on Spaces, Lemma 086D there exists a V-admissible blow up $Y' \to Y$ whose restriction to Y_0 is $Y'_0 \to Y_0$. If $f' : X' \to Y'$ denotes the strict transform of f, then we see what we want is true because f' restricts to an isomorphism over Y'_0. □

Lemma 0.6. *Let S be a scheme. Consider a diagram*

$$\begin{array}{ccccc} X & \longleftarrow & U & \longleftarrow & A \\ {\scriptstyle f}\downarrow & & {\scriptstyle f|_U}\downarrow & & \downarrow \\ Y & \longleftarrow & V & \longleftarrow & B \end{array}$$

of quasi-compact and quasi-separated algebraic spaces over S. Assume

1. *f is proper,*
2. *V is a quasi-compact open of Y, $U = f^{-1}(V)$,*
3. *$B \subset V$ and $A \subset U$ are closed subspaces,*
4. *$f|_A : A \to B$ is an isomorphism, and f is étale at every point of A.*

Then there exists a V-admissible blowing up $Y' \to Y$ such that the strict transform $f' : X' \to Y'$ satisfies: for every geometric point $\overline{a}$ of the closure of $|A|$ in $|X'|$ there exists a quotient $\mathcal{O}_{X',\overline{a}} \to \mathcal{O}$ such that $\mathcal{O}_{Y',f'(\overline{a})} \to \mathcal{O}$ is finite flat.

As you can see from the proof, more is true, but the statement is already long enough and this will be sufficient later on.

Proof. Let $T' \subset |U|$ be the complement of the maximal open on which $f|_U$ is étale. Then T' is closed in $|U|$ and disjoint from $|A|$. Since $|U|$ is a spectral topological space (Properties of Spaces, Lemma 0A4G) we can find constructible closed subsets T_c, T'_c of $|U|$ with $|A| \subset T_c$, $T' \subset T'_c$ such that $T_c \cap T'_c = \emptyset$ (see proof of Lemma 0.5). By Lemma 0.4 there is a U-admissible blowing up $X_1 \to X$ such that T_c and T'_c have disjoint closures in $|X_1|$. Let $X_{1,0}$ be the open subspace of X_1 corresponding to the open $|X_1| \setminus \overline{T}'_c$ and set $U_0 = U \cap X_{1,0}$. Observe that the scheme theoretic image $\overline{A}_1 \subset X_1$ of A is contained in $X_{1,0}$ by construction.

After replacing Y by a V-admissible blowing up and taking strict transforms, we may assume $X_{1,0} \to Y$ is flat, quasi-finite, and of finite presentation, see More on Morphisms of Spaces, Lemmas 087E and 0878. Consider the commutative diagram

$$\begin{array}{ccc} X_1 & \longrightarrow & X \\ & \searrow \quad \swarrow & \\ & Y & \end{array} \qquad \text{and the diagram} \qquad \begin{array}{ccc} \overline{A}_1 & \longrightarrow & \overline{A} \\ & \searrow \quad \swarrow & \\ & \overline{B} & \end{array}$$

of scheme theoretic images. The morphism $\overline{A}_1 \to \overline{A}$ is surjective because it is proper and hence the scheme theoretic image of $\overline{A}_1 \to \overline{A}$ must be equal to $\overline{A}$ and then we can use Morphisms of Spaces, Lemma 0AGD. The statement on étale local rings follows by choosing a lift of the geometric point $\overline{a}$ to a geometric point $\overline{a}_1$ of $\overline{A}_1$ and setting $\mathcal{O} = \mathcal{O}_{X_1,\overline{a}_1}$. Namely, since $X_1 \to Y$ is flat and quasi-finite on $X_{1,0} \supset \overline{A}_1$, the map $\mathcal{O}_{Y',f'(\overline{a})} \to \mathcal{O}_{X_1,\overline{a}_1}$ is finite flat, see Algebra, Lemmas 05WR and 04GG. □

Lemma 0.7. *Let S be a scheme. Let $X \to B$ and $Y \to B$ be morphisms of algebraic spaces over S. Let $U \subset X$ be an open subspace. Let $V \to X \times_B Y$ be a quasi-compact morphism whose composition with the first projection maps into U. Let $Z \subset X \times_B Y$ be the scheme theoretic image of $V \to X \times_B Y$. Let $X' \to X$ be a U-admissible blowup. Then the scheme theoretic image of $V \to X' \times_B Y$ is the strict transform of Z with respect to the blowing up.*

Proof. Denote $Z' \to Z$ the strict transform. The morphism $Z' \to X'$ induces a morphism $Z' \to X' \times_B Y$ which is a closed immersion (as Z' is a closed subspace of $X' \times_X Z$ by definition). Thus to finish the proof it suffices to show that the scheme theoretic image Z'' of $V \to Z'$ is Z'. Observe that $Z'' \subset Z'$ is a closed subspace such that $V \to Z'$ factors through Z''. Since both $V \to X \times_B Y$ and $V \to X' \times_B Y$ are quasi-compact (for the latter this follows from Morphisms of Spaces, Lemma 03KS and the fact that $X' \times_B Y \to X \times_B Y$ is separated as a base change of a proper morphism), by Morphisms of Spaces, Lemma 082Z we see that $Z \cap (U \times_B Y) = Z'' \cap (U \times_B Y)$. Thus the inclusion morphism $Z'' \to Z'$ is an isomorphism away from the exceptional divisor E of $Z' \to Z$. However, the structure sheaf of Z' does not have any nonzero sections supported on E (by definition of strict transforms) and we conclude that the surjection $\mathcal{O}_{Z'} \to \mathcal{O}_{Z''}$ must be an isomorphism. □

Lemma 0.8. *Let S be a scheme. Let B be a quasi-compact and quasi-separated algebraic space over S. Let U be an algebraic space of finite type and separated over B. Let $V \to U$ be an étale morphism. If V has a compactification $V \subset Y$ over B, then there exists a V-admissible blowing up $Y' \to Y$ and an open $V \subset V' \subset Y'$ such that $V \to U$ extends to a proper morphism $V' \to U$.*

Proof. Consider the scheme theoretic image $Z \subset Y \times_B U$ of the "diagonal" morphism $V \to Y \times_B U$. If we replace Y by a V-admissible blowing up, then Z is replaced by the strict transform with respect to this blowing up, see Lemma 0.7. Hence by More on Morphisms of Spaces, Lemma 087F we may assume $Z \to Y$ is an open immersion. If $V' \subset Y$ denotes the image, then we see that the induced morphism $V' \to U$ is proper because the projection $Y \times_B U \to U$ is proper and $V' \cong Z$ is a closed subspace of $Y \times_B U$. □

Proof. Choose a compactification $U_i \subset X_i$ over B for $i = 1, 2$. We may assume U_i is scheme theoretically dense in X_i. We may assume there is an open $V_i \subset X_i$ and a proper morphism $\psi_i : V_i \to U$ extending $U_i \to U$, see Lemma 0.8. Picture

$$\begin{array}{ccccc} U_i & \longrightarrow & V_i & \longrightarrow & X_i \\ \downarrow & \swarrow_{\psi_i} & & & \\ U & & & & \end{array}$$

Denote $Z_1 \subset U$ the reduced closed subspace corresponding to the closed subset $|U| \setminus |U_2|$. Recall that $f^{-1}Z_1$ is a closed subspace of U_1 mapping isomorphically to Z_1. Denote $Z_2 \subset U$ the reduced closed subspace corresponding to the closed subset

$|U| \setminus \mathrm{Im}(|f|) = |U_2| \setminus \mathrm{Im}(|U_1 \times_U U_2| \to |U_2|)$. Thus we have

$$U = U_2 \amalg Z_1 = Z_2 \amalg \mathrm{Im}(f) = Z_2 \amalg \mathrm{Im}(U_1 \times_U U_2 \to U_2) \amalg Z_1$$

set theoretically. Denote $Z_{i,i} \subset V_i$ the inverse image of Z_i under ψ_i. Observe that ψ_2 is an isomorphism over an open neighbourhood of Z_2. Observe that $Z_{1,1} = \psi_1^{-1} Z_1 = f^{-1} Z_1 \amalg T$ for some closed subspace $T \subset V_1$ disjoint from $f^{-1}Z_1$ and furthermore ψ_1 is étale along $f^{-1}Z_1$. Denote $Z_{i,j} \subset V_i$ the inverse image of Z_j under ψ_i. Observe that $\psi_i : Z_{i,j} \to Z_j$ is a proper morphism. Since Z_i and Z_j are disjoint closed subspaces of U, we see that $Z_{i,i}$ and $Z_{i,j}$ are disjoint closed subspaces of V_i.

Denote $\overline{Z}_{i,i}$ and $\overline{Z}_{i,j}$ the scheme theoretic images of $Z_{i,i}$ and $Z_{i,j}$ in X_i. We recall that $|Z_{i,j}|$ is dense in $|\overline{Z}_{i,j}|$, see Morphisms of Spaces, Lemma 088G. After replacing X_i by a V_i-admissible blowup we may assume that $\overline{Z}_{i,i}$ and $\overline{Z}_{i,j}$ are disjoint, see Lemma 0.4. We assume this holds for both X_1 and X_2. Observe that this property is preserved if we replace X_i by a further V_i-admissible blowup. Hence we may replace X_1 by another V_1-admissible blowup and assume $|\overline{Z}_{1,1}|$ is the disjoint union of the closures of $|T|$ and $|f^{-1}Z_1|$ in $|X_1|$.

Set $V_{12} = V_1 \times_U V_2$. We have an immersion $V_{12} \to X_1 \times_B X_2$ which is the composition of the closed immersion $V_{12} = V_1 \times_U V_2 \to V_1 \times_B V_2$ (Morphisms of Spaces, Lemma 03KN) and the open immersion $V_1 \times_B V_2 \to X_1 \times_B X_2$. Let $X_{12} \subset X_1 \times_B X_2$ be the scheme theoretic image of $V_{12} \to X_1 \times_B X_2$. The projection morphisms

$$p_1 : X_{12} \to X_1 \quad \text{and} \quad p_2 : X_{12} \to X_2$$

are proper as X_1 and X_2 are proper over B. If we replace X_1 by a V_1-admissible blowing up, then X_{12} is replaced by the strict transform with respect to this blowing up, see Lemma 0.7.

Denote $\psi : V_{12} \to U$ the compositions $\psi = \psi_1 \circ p_1|_{V_{12}} = \psi_2 \circ p_2|_{V_{12}}$. Consider the closed subspace

$$Z_{12,2} = (p_1|_{V_{12}})^{-1} Z_{1,2} = (p_2|_{V_{12}})^{-1} Z_{2,2} = \psi^{-1} Z_2 \subset V_{12}$$

The morphism $p_1|_{V_{12}} : V_{12} \to V_1$ is an isomorphism over an open neighbourhood of $Z_{1,2}$ because $\psi_2 : V_2 \to U$ is an isomorphism over an open neighbourhood of Z_2 and $V_{12} = V_1 \times_U V_2$. By Lemma 0.5 there exists a V_1-admissible blowing up $X_1' \to X_1$ such that the strict transform $p_1' : X_{12}' \to X_1'$ of p_1 is an isomorphism over an open neighbourhood of the closure of $|Z_{1,2}|$ in $|X_1'|$. After replacing X_1 by X_1' and X_{12} by X_{12}' we may assume that p_1 is an isomorphism over an open neighbourhood of $|\overline{Z}_{1,2}|$.

The result of the previous paragraph tells us that

$$X_{12} \cap (\overline{Z}_{1,2} \times_B \overline{Z}_{2,1}) = \emptyset$$

where the intersection taken in $X_1 \times_B X_2$. Namely, the inverse image $p_1^{-1}\overline{Z}_{1,2}$ in X_{12} maps isomorphically to $\overline{Z}_{1,2}$. In particular, we see that $|Z_{12,2}|$ is dense in $|p_1^{-1}\overline{Z}_{1,2}|$. Thus p_2 maps $|p_1^{-1}\overline{Z}_{1,2}|$ into $|\overline{Z}_{2,2}|$. Since $|\overline{Z}_{2,2}| \cap |\overline{Z}_{2,1}| = \emptyset$ we conclude.

It turns out that we need to do one additional blowing up before we can conclude the argument. Namely, let $V_2 \subset W_2 \subset X_2$ be the open subspace with underlying topological space

$$|W_2| = |V_2| \cup (|X_2| \setminus |\overline{Z}_{2,1}|) = |X_2| \setminus \left(|\overline{Z}_{2,1}| \setminus |Z_{2,1}|\right)$$

Since $p_2(p_1^{-1}\overline{Z}_{1,2})$ is contained in W_2 (see above) we see that replacing X_2 by a W_2-admissible blowup and X_{21} by the corresponding strict transform will preserve the property of p_1 being an isomorphism over an open neighbourhood of $\overline{Z}_{1,2}$. Since $\overline{Z}_{2,1} \cap W_2 = \overline{Z}_{2,1} \cap V_2 = Z_{2,1}$ we see that $Z_{2,1}$ is a closed subspace of W_2 and V_2.

Observe that $V_{12} = V_1 \times_U V_2 = p_1^{-1}(V_1) = p_2^{-1}(V_2)$ as open subspaces of X_{12} as it is the largest open subspace of X_{12} over which the morphism $\psi : V_{12} \to U$ extends; details omitted[1]. We have the following equalities of closed subspaces of V_{12}:

$$p_2^{-1}Z_{2,1} = p_2^{-1}\psi_2^{-1}Z_1 = p_1^{-1}\psi_1^{-1}Z_1 = p_1^{-1}Z_{1,1} = p_1^{-1}f^{-1}Z_1 \amalg p_1^{-1}T$$

Here and below we use the slight abuse of notation of writing p_2 in stead of the restriction of p_2 to V_{12}, etc. Since $p_2^{-1}(Z_{2,1})$ is a closed subspace of $p_2^{-1}(W_2)$ as $Z_{2,1}$ is a closed subspace of W_2 we conclude that also $p_1^{-1}f^{-1}Z_1$ is a closed subspace of $p_2^{-1}(W_2)$. Finally, the morphism $p_2 : X_{12} \to X_2$ is étale at points of $p_1^{-1}f^{-1}Z_1$ as ψ_1 is étale along $f^{-1}Z_1$ and $V_{12} = V_1 \times_U V_2$. Thus we may apply Lemma 0.6 to the morphism $p_2 : X_{12} \to X_2$, the open W_2, the closed subspace $Z_{2,1} \subset W_2$, and the closed subspace $p_1^{-1}f^{-1}Z_1 \subset p_2^{-1}(W_2)$. Hence after replacing X_2 by a W_2-admissible blowup and X_{12} by the corresponding strict transform, we obtain for every geometric point $\overline{y}$ of the closure of $|p_1^{-1}f^{-1}Z_1|$ a local ring map $\mathcal{O}_{X_{12},\overline{y}} \to \mathcal{O}$ such that $\mathcal{O}_{X_2,p_2(\overline{y})} \to \mathcal{O}$ is finite flat.

Consider the algebraic space

$$W_2 = U \coprod\nolimits_{U_2} (X_2 \setminus \overline{Z}_{2,1}),$$

and with $T \subset V_1$ as in the first paragraph the algebraic space

$$W_1 = U \coprod\nolimits_{U_1} (X_1 \setminus \overline{Z}_{1,2} \cup \overline{T}),$$

obtained by pushout, see Lemma 0.2. Let us apply Lemma 0.3 to see that $W_i \to B$ is separated. First, $U \to B$ and $X_i \to B$ are separated. Let us check the quasi-compact immersion $U_i \to U \times_B (X_i \setminus \overline{Z}_{i,j})$ is closed using the valuative criterion, see Morphisms of Spaces, Lemma 03KA. Choose a valuation ring A over B with fraction field K and compatible morphisms $(u, x_i) : \mathrm{Spec}(A) \to U \times_B X_i$ and $u_i : \mathrm{Spec}(K) \to U_i$. Since ψ_i is proper, we can find a unique $v_i : \mathrm{Spec}(A) \to V_i$ compatible with u and u_i. Since X_i is proper over B we see that $x_i = v_i$. If v_i does not factor through $U_i \subset V_i$, then we conclude that x_i maps the closed point of $\mathrm{Spec}(A)$ into $Z_{i,j}$ or T when $i = 1$. This finishes the proof because we removed $\overline{Z}_{i,j}$ and $\overline{T}$ in the construction of W_i.

On the other hand, for any valuation ring A over B with fraction field K and any morphism

$$\gamma : \mathrm{Spec}(K) \to \mathrm{Im}(U_1 \times_U U_2 \to U)$$

over B, we claim that after replacing A by an extension of valuation rings, there is an i and an extension of γ to a morphism $h_i : \mathrm{Spec}(A) \to W_i$. Namely, we first extend γ to a morphism $g_2 : \mathrm{Spec}(A) \to X_2$ using the valuative criterion of properness. If the image of g_2 does not meet $\overline{Z}_{2,1}$, then we obtain our morphism into W_2. Otherwise, denote $\overline{z} \in \overline{Z}_{2,1}$ a geometric point lying over the image of the closed point under g_2. We may lift this to a geometric point $\overline{y}$ of X_{12} in the closure of $|p_1^{-1}f^{-1}Z_1|$ because the map of spaces $|p_1^{-1}f^{-1}Z_1| \to |\overline{Z}_{2,1}|$ is closed with image containing the dense open $|Z_{2,1}|$. After replacing A by its strict henselization (More on Algebra, Lemma 0ASK) we get the following diagram

$$\begin{array}{ccccc} A & \dashrightarrow & & & A' \\ \uparrow & & & & \uparrow \\ \mathcal{O}_{X_2,\overline{z}} & \longrightarrow & \mathcal{O}_{X_{12},\overline{y}} & \longrightarrow & \mathcal{O} \end{array}$$

[1]Namely, $V_1 \times_U V_2$ is proper over U so if ψ extends to a larger open of X_{12}, then $V_1 \times_U V_2$ would be closed in this open by Morphisms of Spaces, Lemma 04NX. Then we get equality as $V_{12} \subset X_{12}$ is dense.

where $\mathcal{O}_{X_{12},\overline{y}} \to \mathcal{O}$ is the map we found in the 5th paragraph of the proof. Since the horizontal composition is finite and flat we can find an extension of valuation rings A'/A and dotted arrow making the diagram commute. After replacing A by A' this means that we obtain a lift $g_{12} : \mathrm{Spec}(A) \to X_{12}$ whose closed point maps into the closure of $|p_1^{-1} f^{-1} Z_1|$. Then $g_1 = p_1 \circ g_{12} : \mathrm{Spec}(A) \to X_1$ is a morphism whose closed point maps into the closure of $|f^{-1} Z_1|$. Since the closure of $|f^{-1} Z_1|$ is disjoint from the closure of $|T|$ and contained in $|\overline{Z}_{1,1}|$ which is disjoint from $|\overline{Z}_{1,2}|$ we conclude that g_1 defines a morphism $h_1 : \mathrm{Spec}(A) \to W_1$ as desired.

Consider a diagram

$$\begin{array}{ccccc} W_1' & \longrightarrow & W & \longleftarrow & W_2' \\ \downarrow & \nwarrow & \uparrow & \nearrow & \downarrow \\ W_1 & \longleftarrow & U & \longrightarrow & W_2 \end{array}$$

as in More on Morphisms of Spaces, Lemma 088Q. By the previous paragraph for every solid diagram

$$\begin{array}{ccc} \mathrm{Spec}(K) & \xrightarrow{\gamma} & W \\ \downarrow & \nearrow & \downarrow \\ \mathrm{Spec}(A) & \longrightarrow & B \end{array}$$

where $\mathrm{Im}(\gamma) \subset \mathrm{Im}(U_1 \times_U U_2 \to U)$ there is an i and an extension $h_i : \mathrm{Spec}(A) \to W_i$ of γ after possibly replacing A by an extension of valuation rings. Using the valuative criterion of properness for $W_i' \to W_i$, we can then lift h_i to $h_i' : \mathrm{Spec}(A) \to W_i'$. Hence the dotted arrow in the diagram exists after possibly extending A. Since W is separated over B, we see that the choice of extension isn't needed and the arrow is unique as well, see Morphisms of Spaces, Lemmas 0A3W and 03KU. Then finally the existence of the dotted arrow implies that $W \to B$ is universally closed by Morphisms of Spaces, Lemma 089G. As $W \to B$ is already of finite type and separated, we win. □

References

[1] Brian Conrad, Max Lieblich, and Martin Olsson. Nagata compactification for algebraic spaces. *J. Inst. Math. Jussieu*, 11(4):747–814, 2012.